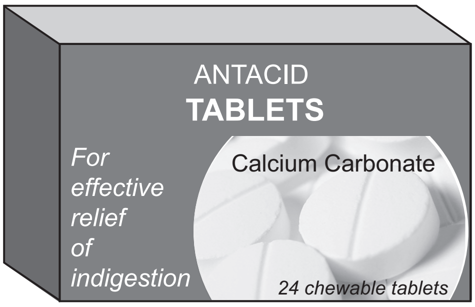
9. Indigestion is caused by excess hydrochloric acid in the stomach. The calcium carbonate in an antacid tablet neutralises the excess acid.

Calcium carbonate reacts with hydrochloric acid according to the equation below.

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$



To determine how much calcium carbonate is present, a crushed indigestion tablet was mixed with water and titrated with dilute hydrochloric acid of concentration 1.0 mol/dm^3. The end-point was determined using the indicator phenolphthalein. The procedure was repeated three times and the mean volume of hydrochloric acid solution needed to neutralise one tablet was found to be 15.2 cm^3.

(a) Calculate the number of moles of hydrochloric acid in 15.2 cm^3 of the 1.0 mol/dm^3 solution. [2]

Number of moles = ... mol

(b) Calculate the number of moles of calcium carbonate in one tablet. [1]

Number of moles = ... mol

Examiner only

(c) Calculate the mass of calcium carbonate in one tablet. Give your answer in **milligrams**, **mg**. [3]

A_r(Ca) = 40 A_r(C) = 12 A_r(O) = 16

Mass of calcium carbonate = ... mg

(d) The diagram shows the labelling on the packet of indigestion tablets.

Each tablet contains:

calcium carbonate	680 mg
magnesium carbonate	80 mg
sucrose and glucose	250 mg
flavouring	
talc	
saccharin	
sodium and magnesium stearate	

Suggest an explanation for the difference between your answer in part *(c)* and the information given on the packet. [1]

....

....

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END OF PAPER